

RESEARCH REPORT

Chronicle's Extraction Phase

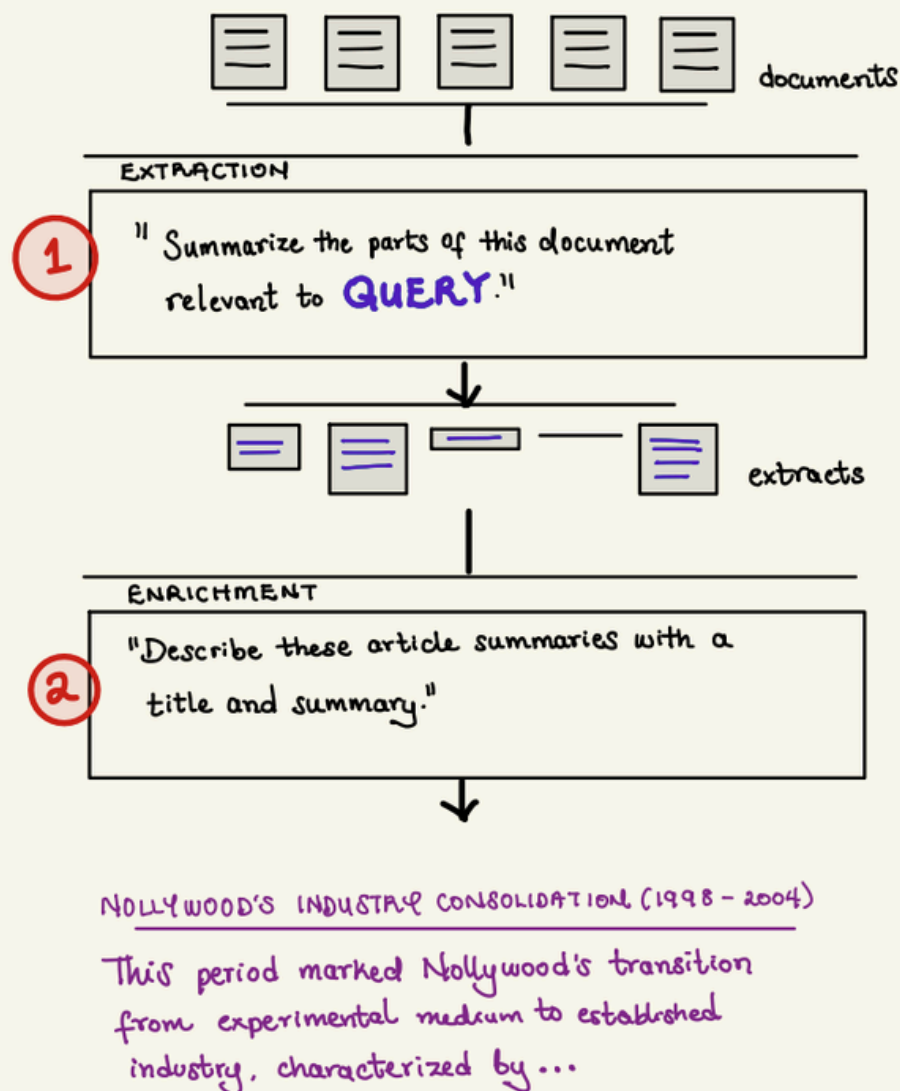
Current Extraction Phase

Chronicle's enrichment pipeline runs in two phases.

Phase 1 (extraction) takes each document returned by search and uses an LLM to distill the portions most relevant to the user's query.

Phase 2 (enrichment) synthesizes those extracts into cluster titles and summaries.

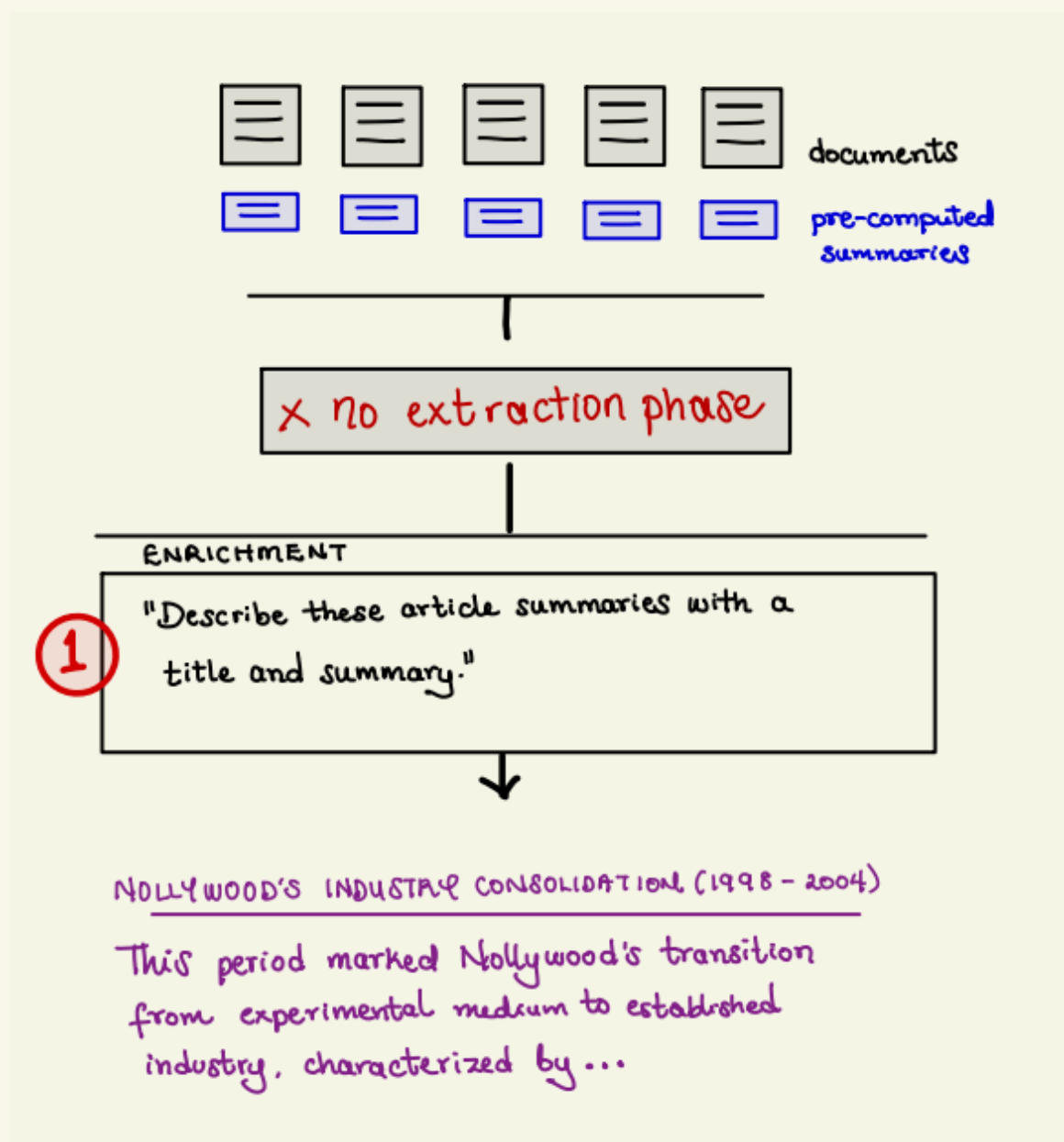
Phase 1 exists because newspaper pages carry multiple full stories, and passing these pages into Phase 2 would flood the context with noise, inflate costs, and slow the pipeline.



But...

The on-the-fly extraction is **slow and expensive**, with some queries taking up to **10 minutes** and costing up to **\$2**. (The original search query on archivi.ng is instantaneous, so this is a considerable degradation in user experience).

This report explores replacing the on-the-fly queries with **pre-computed summaries**.



Current Database Summaries

These were generated for display, not as inputs for further summarization.

Example:

<https://archivi.ng/search/alNY84sBg5yxONaM5LvB>

The Nigerian film industry, also known as Nollywood, has banned eight actors and actresses for alleged unprofessional conduct.

The banned actors and actresses have been accused of refusing to show up on location, ignoring the rules of the industry, and charging outrageous fees.

The ban has sparked controversy and reactions from friends and colleagues of the banned actors, as well as other industry stakeholders.

Some of the banned actors have reportedly begun going cap in hand to beg for forgiveness and refund some of the monies collected from the marketers and producers.

Regenerate Richer Summaries

Example:

<https://archivi.ng/search/alNY84sBg5yxONaM5LvB>

The Association of Movie Marketers and Producers imposed a **one-year** ban on ten prominent Nigerian actors and actresses.

The banned artistes include: **Ramsey Noah, Segun Arinze, Jimlyke, Omotola Jalade-Ekeinde, Richard Mofe-Damijo (RMD), Nkem Owoh, Genevieve Nnaji**, and others. These actors collectively refer to themselves as the **G8**.

Reasons for the Ban:

The ban was imposed due to alleged unprofessional conduct, including ...

Industry Response:

Movie marketers and producers have begun scouting for raw talent on university campuses to replace An anonymous actor stated: **"It serves them right."**

Refunds and Reconciliation Efforts:

According to **Entertainment Cafe** sources, some banned actors have begun refunding monies collected for films not yet shot: Ramsey Noah refunded **N550,000** in cash plus a cheque of **N450,000**. Nkem Owoh refunded **N1.5 million**. Genevieve Nnaji refunded **N1.3 million**. Jim lyke refunded **N3 million**...

Names, Quotes, Figures, Dates, and other details are generally preserved.

Research Question

How would we compare **latency**, **cost** and **quality** between these three extraction options?

On-The-Fly

- Query-Specific, On-the-Fly extraction (Current Implementation).

Vanilla Summaries

- Query-agnostic, Current Pre-Generated Summaries from the database.

Rich Summaries

- Query-agnostic, Richer Pre-Generated Summaries specially for Chronicle.

Latency (seconds)

	 VANILLA	RICHER	ON THE FLY
QUERY 1	49	57	170
QUERY 2	29	33	154
QUERY 3	30	39	139

Cost (dollars)

	 VANILLA	RICHER	ON THE FLY
QUERY 1	0.05	0.15	0.38
QUERY 2	0.03	0.07	0.32
QUERY 3	0.03	0.06	0.28

Note: The cost of the richer summaries does not include the cost to regenerate them, just the cost per query.

Notes on Quality

None are “that bad”

Query: “Jay-Jay Okocha” (1994-97)

ON THE FLY

- enugu rangers -> european stardom + olympic glory
- austin 'jay-jay' okocha
- midfield maestro
- 9 league matches for enugu rangers
- moved to Germany in 1990
- signed with Eintracht Frankfurt in January 1992
- praise from World Cup captain Lothar Matthäus
- individual to team play
- 'the days of one man show are over.'
- **Nigerian gold medal at Atlanta '96**
- **crucial penalty against Japan**
- Frankfurt, Bundesliga -> Turkish Fenerbahce 1996 due to relegation
- expressed frustration with lower quality of Turkish football
- **Nigeria qualified for World Cup for France 1998**
- he + akpborie recommended their former Frankfurt coach Dragoslav Stepanovic to Nigeria in 1997

VANILLA SUMMARY

- austin 'jay-jay' okocha
- enugu -> international stardom + turkish league Discontent
- national grammar school in Kano
- 1996 Atlanta Olympics qualifiers in Zimbabwe
- midfield maestro
- German -> Turkish side Fenerbahçe
- complained about Turkish team
- **Ochuko Utovbivbi, emerging talent compared to Okocha**

Unimportant detail gleaned from an article primarily about Ochuko. Jay-Jay Okocha was very briefly mentioned.



- DETAILS UNIQUE TO THIS RESULT
- IMPORTANT DETAILS UNIQUE TO THIS RESULT
- ERROR

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RICH SUMMARY

- austin 'jay-jay' okocha
- bundesliga breakthrough -> turkish disappointment
- midfield maestro
- Eintracht Frankfurt in Germany's Bundesliga (signed in January 1992)
- 1996 Atlanta Olympics, a penalty against Japan
- 1997 Frankfurt -> Turkish club Fenerbahçe (controversial, mixed results)
- expressed disappointment on an interview
- stated plans to leave in 1998
- 'never go backward' to second-division football
- Nigerian football: World Cup qualifiers against teams like Burkina Faso and Kenya
- former coach Dragoslav Stepanovic as a Super Eagles coach recommendation

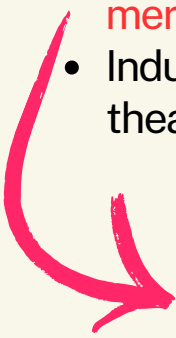
Query: “Nollywood” (1991-98)

ON THE FLY

- 1991, NEK Video Links by Kenneth Nnebue direct-to-home
- **U-Matic format and VHS cassettes**
- First release (Aje Ni Iyami, March 1991) great response
- **8 additional films by November.**
- Started Yoruba but expanded across ethnic groups
- **By the mid-1990s, economic downturn caused celluloid (6-8 million) -> affordable video production**
- Rapid profitization affected quality
- **Ego Boyo, Dr Ola Makinde (industry figures) complained about rushed prods, poor actor pay, fetish films**
- **Jide Kosoko continued to create despite econ downturn**
- **1996 Nigerian Motion Pictures Association organized guilds of tech, prod, actors**
- **National Film and Video Censors board regulated flood of new releases**

VANILLA SUMMARY

- Traditional theater -> Home-video industry (nollywood)
- Kenneth Nnebue and NEK Video Links spearheaded revolution
- Marketing indigenous films directly to consumers
- beginning with Yoruba prods and expanding to English versions
- **Ola Makinwa showed Nigerian films in London's Hackney Empire**
- **advocated for govt support and better distribution.**
- **voiced concerns about tech quality, language barriers, artistic merit < profit.**
- Industry veterans like Jide Kosoko and **Ego Boyo** navigated trad theater and new video format



HALLUCINATION

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RICH SUMMARY

- home video film as the origin of Nollywood
- start: March 1991 when Kenneth Nnebue's NEK Video Links released 'Aje Ni Iyami'
- direct to consumer marketing
- U-matic format, VHS tapes
- 9 films by November 1991
- Producer Jide Kosoko - econ downturn made celluloid (6-8M) too costly.
- Actress Ego Boyo called for better quality.
- 1997 - Industry observers alarmed by fetishistic films
- Ifeanyi Dike of the Nigerian Actors Guild talked about international image about the cultish films.

Hallucinations

Hallucinations show up a few times in all of the methods.

Examples

- **On-the-fly summarization:** In one instance, this method produces a claim that Jay-Jay Okocha became a Turkish citizen during his time at Fenerbahçe. While this is true, it is not mentioned in any of the documents the summary was based on.
- **Rich pre-computed summary:** In one instance, this claimed Hull City was in England's Premier League. At the time, this was not the case. And this claim is not supported by any of the underlying documents.

Notes

Sending smaller inputs to the enrichment LLM seems to reduce hallucinations.

Hallucinations come in highly plausible forms like exaggerating an opinion. In one example, “terrible performance at the 2010 Olympics” was presented as “their worst ever Olympic performance.”

Other Issues

Obvious gossip and sensational sources are generally ignored by the LLM. But some opinion pieces are included as fact.

For example, claiming that Nigerian women in politics were “satisfied with mobilization roles” in lieu of seeking office. While this opinion appears in a source, it is incorrectly presented as a fact in a final summary.